Shrinking Equivalence Radii and Vanishing Borrowing in Distributional Test-then-pool:

Consistency, Selection Stability and the Price in Local Power

Builds on	L. Yang, X. Liu and R. J. Evans (2026), *Data Fusion with Distributional Equivalence Test-then-pool*, arXiv:2603.11867v2, Remark 3.19.
Question	Can a radius $\theta_n \to 0$ or a borrowing mass $b_n \to 0$, fixed before seeing data, remove the power blind spot of fixed-radius equivalence test-then-pool, and at what cost?
Content	Selection asymptotics with shrinking radii; exact and asymptotic calibration; a rate-sharp local blind spot; vanishing borrowing; partial-bootstrap degeneracy; Monte Carlo study with sample-size-dependent populations and an independent-seed confirmation.
Status	Research draft, 18 October 2026. Proofs are the author's own and have not been independently refereed. Prepared with AI assistance (Claude); ideas from an earlier AI draft (GPT, 17 October 2026) are credited where reused.
Companion	Study A: *Multi-source distributional test-then-pool*.
Materials	Code, seeds, raw results: `https://github.com/Yu-Jusang/ttp-extensions`

Abstract

With a fixed equivalence radius θ, the distributional test-then-pool (TTP) procedure of Yang et al. (2026) keeps borrowing a historical control that differs from the current control by up to θ, and it is not consistent against alternatives in which the treatment effect points in the direction of the historical shift (their Remark 3.19). We analyse prespecified schedules $\theta_n = \theta_0(n/n_0)^{-a}$ and borrowing masses $b_n \to 0$. For the fusion stage we prove that an identical source is retained with probability tending to one when $n\theta_n^2 \to \infty$, and is eventually discarded when $\sqrt{n}\,\theta_n \to 0$; a fixed shifted source is discarded whenever $\theta_n \to 0$; we also give explicit selection-error bounds. Calibrated by full reselection, the final test has finite-sample level for every schedule, and it is consistent against every fixed alternative as soon as $\theta_n \to 0$. Freezing the selection, as in the original procedure, gives pointwise asymptotic validity and consistency by coupling with an oracle, but not uniformly. We prove that the price of shrinking radii is a local blind spot of exact order θ_n: there are alternatives with $D(Q_c, Q_t) = c\,\theta_n \gg n^{-1/2}$ for which the power tends to zero although the current-only test has power tending to one. No radius schedule both retains identical sources and is uniformly consistent at the $\sqrt{n}$ scale. A minP safeguard, or the soft risk weights of the companion study, avoids this trade-off. Vanishing borrowing mass restores consistency but gives no first-order local power gain from identical sources. The original partial bootstrap degenerates at $Q_h = Q_c$; a Gaussian-limit calculation and simulations suggest that it is conservative there, which we state as a conjecture. Simulations with n up to 800 confirm these patterns and show their finite-sample sizes.

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1 Introduction

Equivalence test-then-pool (Li et al., 2020; Yang et al., 2026) borrows a historical control sample only after an equivalence test has shown it to be within distance θ of the current control. Equivalence is not identity. A merged source can differ from the current control by almost θ, and this displacement can cancel a real treatment effect. Yang et al. (2026, Theorems 3.6, 3.9, Corollary 3.18) show that their procedure is consistent only when the treatment contrast is large relative to the projected historical shift. In their Remark 3.19 they suggest letting the radius depend on the sample size, $\theta = \theta_n = o(1)$, and leave the analysis open. This paper takes up that analysis.

We study prespecified, deterministic schedules,

$$\theta_n = \theta_0\Big(\frac{n}{n_0}\Big)^{-a}, \qquad b_n = b_0\Big(\frac{n}{n_0}\Big)^{-b},$$

fixed before any outcome is seen. Data-driven tuning of θ is a different problem and would need its own selection correction (Section 3.5 of Yang et al., 2026). We distinguish three design choices:

1. *what shrinks*: the tolerated discrepancy of a borrowed source (θ_n) or the total weight given to historical data (b_n);
2. *how the final test is calibrated*: by freezing the observed fusion decision (original architecture) or by re-running fusion in every permutation (full reselection, as in the companion Study A);
3. *which causal-stage test is used*: partial permutation or the original partial bootstrap.

Answers in brief.

- *Consistency against all fixed alternatives.* With full reselection it holds whenever $\theta_n \to 0$, at any rate (Theorem 3.6). With frozen selection it holds pointwise under $\theta_n \to 0$ and $n\theta_n^2 \to \infty$ (Theorem 3.10).
- *Keeping an identical source.* This needs θ_n to shrink more slowly than the sampling error: $n\theta_n^2 \to \infty$ suffices, and $\sqrt{n}\,\theta_n \to 0$ loses the source (Theorem 3.3). In the power family, $0 < a < 1/2$ suffices, and $a > 1/2$ discards identical sources.
- *Too fast or too slow.* Too fast forfeits borrowing. Too slow leaves a blind spot that persists over a long range of n. Even the ideal rate leaves a *local* blind spot of size $\asymp \theta_n$ (Theorem 3.7), so a radius schedule cannot have both properties (Corollary 3.8).
- *Constants.* We give a planning rule for (θ_0, n_0, a) in Section 3.8; it is not an optimality result.
- *Local and sample-size-dependent discrepancies.* A source at distance $d_{h,n}$ is kept when $\sqrt{n}(\theta_n - d_{h,n}) \to \infty$ and dropped when $\sqrt{n}(d_{h,n} - \theta_n) \to \infty$; near the boundary the selection is random, and frozen calibration can be distorted.

We separate pointwise results (fixed distributions) from uniform ones (distributions that may change with n), and partial-permutation results from partial-bootstrap ones, whose degenerate limits require separate treatment (Section 3.7). We never call a rate optimal: all rate statements are sufficient conditions, matching counterexamples, or experimental choices.

2 Background

2.1 Setting and notation

The setting is that of Yang et al. (2026). We observe independent samples Q_c^m (current control), Q_t^n (treatment) and $Q_{h_j}^{\ell_j}$ ($j = 1, \dots, J$; $J = 1$ in the original), with fixed ratios $m/n \to c_1 > 0$ and $\ell_j/n \to c_{2j} > 0$ (their Assumption 3.2), and test $H_0 : Q_c = Q_t$. As in the original, $\gamma = m/(m+\ell)$ and $\eta = n/(m+n)$; we also write $\lambda = m/N = 1-\eta$, $N = m+n$.

Assumption 2.1 (Kernel)**.** k is a bounded, measurable, characteristic kernel with separable RKHS $\mathcal{H}_k$ and $\kappa^2 = \sup_x k(x,x) < \infty$.

ξ_R is the kernel mean embedding of R, and $\hat{\xi}$ denotes an empirical embedding. $D(P,Q) = \|\xi_P - \xi_Q\|$ **is the MMD, not its square**; test statistics are squared MMDs. For a set S of sources pooled by sample size,

$$Q_{f,S}^{m+\ell_S} = \frac{m}{m+\ell_S} Q_c^m + \sum_{j \in S} \frac{\ell_j}{m+\ell_S} Q_{h_j}^{\ell_j}, \qquad \ell_S = \sum_{j \in S} \ell_j.$$

Constants: $v_{m,n} = \kappa^2(1/m + 1/n)$, $M_{m,n} = \kappa^2 N^2/\{mn(N-1)\}$, $L = B+1$ (permutations including the observed allocation), and $k_\alpha = \lfloor \alpha L \rfloor$.

2.2 The original procedure and its open question

Yang et al. (2026) test $H_0^f : D(Q_c, Q_h) \geq \theta$ with statistic $\theta - D(Q_c^m, Q_h^\ell)$ and a bootstrap quantile q of $D(Q_c^m, Q_c) + D(Q_h^\ell, Q_h)$ (Liu and Gandy, 2026, Section 4.2), merging when $D(Q_c^m, Q_h^\ell) + q < \theta$ (their Section 3.1, Theorem 3.1). After merging, they use either the partial bootstrap statistic $\Delta = \sqrt{n}\{D^2(Q_f^{m+\ell}, Q_t^n) - D^2(Q_f^{m+\ell}, Q_c^m)\}$ (Algorithm 2), which is asymptotically valid for $Q_h \neq Q_c$ (Theorem 3.5), or the partial permutation statistic $D^2(Q_f^{m+\ell}, Q_t^n)$ with only current labels permuted (Theorem 3.8). Write $A = (1-\gamma)(\xi_h - \xi_c)$ and $\Gamma = \xi_t - \xi_c$. The consistency arguments in the original proofs (Appendices B.4 and B.6) compare population limits of the observed and resampled statistics, and the resulting gaps are

$$g_{\mathrm{PB}} = \|\Gamma\|^2 - 2\langle A, \Gamma \rangle, \qquad g_{\mathrm{PP}} = (1-r)\big\{(1+r)\|\Gamma\|^2 - 2\langle A, \Gamma \rangle\big\}, \quad r = (1-\gamma)\eta.$$

The test is consistent when the gap is positive. The partial-bootstrap condition is exactly the printed $2(1-\gamma)D(Q_h, Q_c)\cos\beta < D(Q_c, Q_t)$ of Theorem 3.6. For partial permutation we re-derived the gap from the limits $\|A - \Gamma\|^2$ (observed) and $\|A - r\Gamma\|^2$ (permuted, with the pooled current mixture $\lambda Q_c + \eta Q_t$). This gives $2(1-\gamma)D(Q_h, Q_c)\cos\beta < \{1 + (1-\gamma)\eta\}D(Q_c, Q_t)$, which differs in form from the condition printed in Theorem 3.9; we could not reconcile the two parameterisations, and nothing below depends on the difference. Both gaps can be nonpositive when $\xi_t - \xi_c$ points along $\xi_h - \xi_c$, and with a fixed θ such a source is merged with probability tending to one. Remark 3.19 therefore records a lack of consistency against all alternatives.

The companion Study A reviews two further steps of the original argument: conditioning the permutation test on the fusion event (their Theorem A.4) and the boundary limit of the fusion test. It shows that freezing a selection is controlled in general only by $|\mathcal{D}|\alpha$, and that full reselection is exact. We reuse these results and refer to them as A-Theorem 3.2 (exact level), A-Theorem 3.12 (Type-II bound) and A-Proposition 3.16 (minP safeguard). Their statements are repeated where needed.

2.3 Related work

Shrinking radii are pretest thresholds that vanish with n, in the tradition of Hodges-type supereffi-ciency. Post-model-selection estimators with consistent selection behave like the oracle pointwise but not uniformly over parameter sequences, and their finite-sample distributions cannot be estimated uniformly consistently (Leeb and Pötscher, 2005). Our frozen-selection results are a testing analogue of that phenomenon. Yang et al. (2023) analyse a test-based integration estimator under local alternatives and choose the pretest threshold adaptively by minimising mean squared error. Zhu et al. (2025) achieve exact Type-I control under selective borrowing by randomization inference. Local and minimax power of permutation and MMD tests is studied by Kim et al. (2022), and the $n^{-1/2}$ minimax rate for embedding estimation by Tolstikhin et al. (2017). Optimal equivalence tests under local asymptotics are treated by Romano (2005). Bayesian dynamic borrowing (Ibrahim and Chen, 2000; Hobbs et al., 2011; Schmidli et al., 2014; Kaizer et al., 2018; Viele et al., 2014) controls borrowing through prior discounting rather than prespecified schedules.

3 Method

3.1 Procedures

Fix a schedule (θ_n), fusion levels $\alpha_{f,j}$ (Bonferroni α_f/J by default), B and B_f.

Frozen radius test F-θ_n (original architecture). For each j, compute q_j by Efron's bootstrap from the observed current control and from source j, accept j if $D(Q_c^m, Q_{h_j}^{\ell_j}) + q_j < \theta_n$, pool the accepted set $\hat{S}$ by sample size, and run partial permutation with $\hat{S}$ held fixed. If $\hat{S} = \emptyset$ this is the current-only permutation test. PB-θ_n uses the original partial bootstrap instead of partial permutation.

Reselected radius test R-θ_n. In every permutation row b (allocation A_b), recompute

$$\hat{S}_b = \{j : D(Q_{c,A_b}^m, Q_{h_j}^{\ell_j}) + q_j < \theta_n\}, \qquad S_b = D^2\big(Q_{f,\hat{S}_b}^{m+\ell_{\hat{S}_b}}(A_b),\, Q_{t,A_b}^n\big),$$

and use $p = L^{-1}\#\{b : S_b \geq S_0\}$. To keep the calibration exact and cheap we use a *label-invariant* critical value: q_j is the $\lceil (1-\alpha_{f,j})B_f \rceil$-th order statistic of

$$D(Q_{\text{pool}}^{*m}, Q_{\text{pool}}^N) + D(Q_{h_j}^{*\ell_j}, Q_{h_j}^{\ell_j}),$$

where Q_{pool}^{*m} is drawn with replacement from the whole current pool. It is computed once and used in all rows. Under H_0 this targets the same law as the original bootstrap, since the pool is i.i.d. from Q_c. Under H_1 it resamples from the pooled current mixture, which changes the fusion error rates but not the final level.

Vanishing mass V-b_n. Fixed weights $w_{n,j} = b_n/J$ on every source, with partial permutation. The rule involves no selection.

Safeguard G. minP combination of the current-only statistic and R-θ_n over the same rows (A-Proposition 3.16).

3.2 Calibration: exact for reselection, pointwise for freezing

Proposition 3.1 (Exact level for every schedule)**.** *For any deterministic sequence (θ_n), any J, any historical distributions and any B, R-θ_n, V-b_n and G satisfy $\mathbb{P}_{H_0}(\text{reject}) \le \lfloor \alpha L \rfloor / L$.*

The label-invariant q_j and the historical bootstrap weights are functions of the unordered pool and of independent auxiliary randomness, so the row statistic is a symmetric function of the control values (A-Assumption 3.1). The result then follows from A-Theorem 3.2.

For frozen selection, A-Proposition 3.3 gives the uniform bound $\mathbb{P}_{H_0}(\text{reject}) \le \min(1, 2^J \alpha)$, which holds for every n and every sequence of distributions. Better bounds need the selection to stabilise.

3.3 Fusion with a shrinking radius

Lemma 3.2 (Order of the fusion critical value)**.** *For either bootstrap variant, with $k_f = \lceil (1 - \alpha_{f,j}) B_f \rceil$, $\mathbb{E} q_j^2 \le \bar{q}_{n,j}^2 := 2\kappa^2 (1/m + 1/\ell_j) B_f / (B_f - k_f + 1)$.*

Theorem 3.3 (Selection with a shrinking radius)**.** *Let $\hat{D}_j = D(Q_c^m, Q_{h_j}^{\ell_j})$, $d_j = D(Q_c, Q_{h_j})$ and $s_j = \kappa^2 (1/m + 1/\ell_j)$. For every n, every row (with Q_c^m replaced by the row's control sample in the reselected test) and all distributions:*

(a) if $d_j < \theta_n$, then $\mathbb{P}(j \notin \hat{S}) \le 4(s_j + \bar{q}_{n,j}^2)/(\theta_n - d_j)^2$;

(b) if $d_j > \theta_n$, then $\mathbb{P}(j \in \hat{S}) \le s_j / (d_j - \theta_n)^2$.

Consequently, for fixed distributions with $\theta_n \to 0$: a source with $d_j > 0$ is discarded with probability tending to one; an identical source ($d_j = 0$) is retained with probability tending to one if $n\theta_n^2 \to \infty$; and if $\sqrt{n}\,\theta_n \to 0$ and $v_{Q_c} > 0$, an identical source is discarded with probability tending to one. For sample-size-dependent discrepancies $d_j = d_{j,n}$, the source is retained with probability tending to one if $\sqrt{n}(\theta_n - d_{j,n}) \to \infty$ and discarded if $\sqrt{n}(d_{j,n} - \theta_n) \to \infty$, uniformly over all distributions satisfying the respective condition.

Corollary 3.4 (Stabilisation to the oracle set)**.** *Let $S_0 = \{j : Q_{h_j} = Q_c\}$, $\theta_n \to 0$ and $n\theta_n^2 \to \infty$. For fixed distributions, $\mathbb{P}(\hat{S} \ne S_0) \le e_n$ with*

$$e_n = \sum_{j \notin S_0} \frac{s_j}{(d_j - \theta_n)^2} + \sum_{j \in S_0} \frac{4(s_j + \bar{q}_{n,j}^2)}{\theta_n^2} \to 0$$

for all n large enough that $\theta_n < \min_{j \notin S_0} d_j$.

Remark 3.5 (The boundary rate $a = 1/2$). If $\sqrt{n}\,\theta_n \to c \in (0, \infty)$ and $Q_{h_j} = Q_c$, then $\sqrt{n}\hat{D}_j$ converges to the norm of a nondegenerate Gaussian element and $\sqrt{n}\,q_j$ to a random limit, so the acceptance probability converges, under a bootstrap-consistency condition, to a constant strictly between 0 and 1. We do not prove this limit; the experiments show intermediate acceptance rates for $a = 0.45$ at moderate n.

3.4 Reselected shrinking radius: consistency and its price

Let $\rho = \ell_{\text{tot}}/(m + \ell_{\text{tot}})$ with $\ell_{\text{tot}} = \sum_j \ell_j$.

Theorem 3.6 (R-θ_n: Type-II bound and consistency)**.** *For every n and all distributions with $d = D(Q_c, Q_t) > 2\rho\theta_n$, if $k_\alpha \geq 1$,*

$$\mathbb{P}(\text{R-}\theta_n \text{ does not reject}) \leq \frac{4(v_{m,n} + BM_{m,n}/k_\alpha)}{(d - 2\rho\theta_n)^2}.$$

Hence $\theta_n \to 0$ (any rate) gives consistency against every fixed alternative, and uniform consistency over $\{D(Q_c, Q_t) \geq d_n\}$, for all historical distributions, whenever $\sqrt{n}(d_n - 2\rho\theta_n) \to \infty$.

The detection boundary in Theorem 3.6 is of order θ_n, larger than the current-only boundary $n^{-1/2}$ when $\sqrt{n}\theta_n \to \infty$. The next theorem shows that this is not a weakness of the proof.

Theorem 3.7 (Local blind spot of exact order θ_n)**.** *Let $J = 1$, fixed ratios, B and B_f fixed with $k_\alpha \geq 1$, $\theta_n \to 0$ and $\sqrt{n}\theta_n \to \infty$. Fix $c \in (0, 1)$ and let $Q_{h,n}$ be any sequence with $D(Q_c, Q_{h,n}) = c\,\theta_n$. Define the local alternative*

$$Q_{t,n} = \gamma Q_c + (1 - \gamma)Q_{h,n}, \qquad D(Q_c, Q_{t,n}) = (1 - \gamma)c\,\theta_n.$$

Then $\sqrt{n}\,D(Q_c, Q_{t,n}) \to \infty$ and the current-only test and G have power tending to one, but the power of R-θ_n and of F-θ_n tends to zero.

Corollary 3.8 (No radius schedule has both properties)**.** *Let (θ_n) be any deterministic sequence with $\theta_n \to 0$, and use R-θ_n or F-θ_n with $J = 1$ and fixed B, B_f. If identical sources are retained with probability bounded away from zero along a subsequence, then $\sqrt{n}\,\theta_n \not\to 0$ along it (Theorem 3.3). If $\sqrt{n}\,\theta_n \to \infty$, Theorem 3.7 gives alternatives with $\sqrt{n}\,D(Q_c, Q_{t,n}) \to \infty$ and power tending to zero. So a shrinking radius cannot both retain identical sources with probability tending to one and be uniformly consistent over $\{\sqrt{n}\,D(Q_c, Q_t) \to \infty\}$.*

Remark 3.9 (Escaping the trade-off). The safeguard G is uniformly consistent at the $\sqrt{n}$ scale (A-Proposition 3.16) and can retain identical sources through its borrowing component, at the cost of testing each component at about level $\alpha/2$. The soft risk weights of Study A (A-Theorem 3.6, A-Corollary 3.13) give an identical source a weight that does not vanish in probability, because $m\hat{D}^2/\hat{v}_c$ has a nondegenerate limit. They also move the fused control by at most $O(n^{-1/2})$ deterministically, so they too escape Corollary 3.8. Hard thresholding at θ_n cannot do both, because accepted sources may be displaced by up to $\theta_n \gg n^{-1/2}$.

3.5 Frozen shrinking radius: oracle coupling

The following result and its coupling proof are adapted from the earlier AI draft (its Theorem 3.8). We restate it with explicit constants and add the non-uniformity discussion.

Theorem 3.10 (Frozen selection: pointwise validity and consistency)**.** *Fix the distributions, J, the ratios, B with $k_\alpha \geq 1$, and B_f. Let $\theta_n \to 0$ with $n\theta_n^2 \to \infty$, and let ϕ^{or} be the fixed-rule partial-permutation test that pools exactly $S_0 = \{j : Q_{h_j} = Q_c\}$, computed with the same permutations. Then $|\mathbb{P}(\text{F-}\theta_n \text{ rejects}) - \mathbb{P}(\phi^{\text{or}} = 1)| \leq e_n \to 0$. Hence $\limsup_n \mathbb{P}_{H_0}(\text{F-}\theta_n \text{ rejects}) \leq \alpha$, and F-$\theta_n$ is consistent against every fixed alternative.*

The coupling bound e_n depends on $\min_{j \notin S_0} d_j$ and is not uniform. If a source sits at distance $d_{j,n} \approx \theta_n$ (or $d_{j,n} \approx c\theta_n$ with $\sqrt{n}\theta_n$ only moderately large), then $\mathbb{P}(j \in \hat{S})$ stays away from 0 and 1, the coupling fails, and the selection and the final statistic share the control sample. The only general bound is then $\min(1, 2^J\alpha)$. We check this boundary regime by simulation (BoundNull in Section 4).

3.6 Vanishing borrowing mass

Proposition 3.11 (Vanishing mass). *Let $w_n \in \mathcal{W}_J$ be deterministic with $s(w_n) \le b_n$. V-b_n has exact level. In every row, $D(\hat{Q}_{f,b}, Q^m_{c,A_b}) \le b_n \max_j D(Q^m_{c,A_b}, Q^{\ell_j}_{h_j}) \le 2\kappa b_n$, so A-Theorem 3.12 gives consistency against every fixed alternative if $b_n \to 0$, and uniform consistency over $\{D(Q_c, Q_t) \ge d_n\}$ if $\sqrt{n}(d_n - 4\kappa b_n) \to \infty$.*

Proposition 3.12 (No first-order gain from vanishing mass). *Suppose $D(Q_c, Q_{t,n}) = O(n^{-1/2})$ and $b_n \to 0$. In every row,*

$$n\left|S^{\mathrm{bor}}_b - S^{\mathrm{cur}}_b\right| \le 2\sqrt{n}\|u_b\| \cdot \sqrt{nS^{\mathrm{cur}}_b} + n\|u_b\|^2, \qquad u_b = \sum_j w_{n,j}(\hat{\xi}_j - \hat{\xi}_{c,A_b}).$$

If every source equals Q_c, then $\sqrt{n}\|u_b\| = O_P(b_n)$, so $n(S^{\mathrm{bor}}_b - S^{\mathrm{cur}}_b) \to 0$ in probability for every row. Borrowing with vanishing mass then changes the n-scaled statistics by $o_P(1)$ and yields no first-order local power gain. If a source has a fixed distance $d_j > 0$, the same holds if $\sqrt{n}\, b_n \to 0$. For $b_n \asymp n^{-1/2}$ the borrowed statistic carries a local bias that can raise or lower local power.

By contrast, a shrinking radius with $n\theta_n^2 \to \infty$ keeps an identical source at its full weight $\ell/(m+\ell)$, which is why radius schedules can improve local power when the sources really are identical (Section 5).

3.7 The partial bootstrap under a shrinking radius

With $\theta_n \to 0$ the merged sources are, asymptotically, exactly the identical ones (Corollary 3.4). This is the case $Q_h = Q_c$ that Yang et al. (2026, Theorem 3.5) exclude, so the partial bootstrap needs separate treatment.

Degeneracy. Write $A_n = (1-\gamma)(\hat{\xi}_h - \hat{\xi}_c)$ and $B_n = \hat{\xi}_t - \hat{\xi}_c$, so that $\Delta/\sqrt{n} = -2\langle A_n, B_n\rangle + \|B_n\|^2$. When $Q_h = Q_c = Q_t$, both A_n and B_n are $O_P(n^{-1/2})$, so $\Delta = O_P(n^{-1/2})$ and the relevant scale is $\Lambda := \sqrt{n}\Delta = n\{-2\langle A_n, B_n\rangle + \|B_n\|^2\}$. By the Hilbert-space central limit theorem, $\Lambda \rightsquigarrow -2(1-\gamma)\langle U, V\rangle + \|V\|^2$ with $U = G_h/\sqrt{c_2} - G_c/\sqrt{c_1}$ and $V = G_t - G_c/\sqrt{c_1}$, where G_c, G_t, G_h are i.i.d. centred Gaussian elements with the covariance operator of $k(\cdot, X)$, $X \sim Q_c$. This is a Gaussian chaos of order two, not a normal law.

The bootstrap world differs. In Algorithm 2 of Yang et al. (2026), the historical sample is resampled from Q^{ℓ}_h and the current control from Q^m_c. The bootstrap population therefore has a fixed displacement $A^{\mathrm{obs}} = (1-\gamma)(\hat{\xi}_h - \hat{\xi}_c) \ne 0$ of order $n^{-1/2}$, and conditionally

$$\Lambda^* \rightsquigarrow -2(1-\gamma)\langle U^{\mathrm{obs}} + U^*, V^*\rangle + \|V^*\|^2,$$

where U^{obs} is the realised value of U and (U^*, V^*) is an independent copy of (U, V). The bootstrap law is a mixture indexed by U^{obs}, with an extra term that is absent from the true limit. Validity therefore does not follow from Proposition 3.4 of the original, and we could not prove it.

Conjecture 3.13 (Conservativeness at degeneracy). *Under $Q_h = Q_c = Q_t$ and fixed ratios, $\limsup_n \mathbb{P}(\Delta > \hat{q}^*_{1-\alpha}) \le \alpha$ for the partial bootstrap with $B \to \infty$.*

Evidence. (i) In the scalar (linear-kernel) version of the two limits, Monte Carlo evaluation (2000 draws of U^{obs}, 4000 bootstrap draws each; script `pb_scalar.py`) gives limiting rejection probabilities below 0.05 in all five ratio configurations tried:

Table 1: Summary of guarantees for a single source ($J = 1$), fixed B with $\alpha L \geq 1$. "Pointwise": fixed distributions; "uniform": worst case over sequences.

Procedure	Level	Consistency (fixed $Q_t \neq Q_c$)	Uniform detection boundary
CUR	exact	yes	$\sqrt{n}d_n \to \infty$
F-θ, PB-θ (fixed θ)	asymptotic (original), $\leq 2\alpha$ uniformly	no (Remark 3.19)	none
R-θ (fixed θ)	exact	no (A-Theorem 3.15)	$d_n > 2\rho\theta$
R-θ_n, $\theta_n \to 0$	exact	yes, any rate	$\asymp \theta_n$ (sharp)
F-θ_n, $n\theta_n^2 \to \infty$	pointwise asymptotic; $\leq 2\alpha$ uniformly	yes (pointwise)	$\asymp \theta_n$ (sharp)
PB-θ_n	open (Conjecture 3.13)	not established	–
V-b_n, $b_n \to 0$	exact	yes	$\sqrt{n}(d_n - 4\kappa b_n) \to \infty$
G (minP)	exact	yes	$\sqrt{n}d_n \to \infty$

(c_1, c_2)	$(0.5, 1)$	$(0.5, 0.5)$	$(1, 1)$	$(0.25, 1)$	$(1, 4)$
rejection prob. (MC s.e. ≈ 0.004)	0.033	0.036	0.039	0.029	0.031

(ii) The RBF simulations in Section 5 report PB-ALL under NullIdent. (iii) Yang et al. (2026) report conservative behaviour in their Figure 2. A conservative bootstrap would protect level but lose power exactly where shrinking radii borrow, namely from identical sources.

3.8 Choosing the schedule before the study

The theory restricts the exponent to $0 < a < 1/2$ if identical sources are to be retained. Within that range, smaller a retains sources longer but leaves the blind spot (3.7) wider, of order n^{-a}. A transparent prespecification is the following.

1. Choose a planning sample size n_0 and a tolerance θ_0 at n_0, for instance the largest discrepancy that subject-matter experts would ignore. Alternatively, choose θ_0 so that an identical source is accepted with a target probability π_0 at n_0 under a planning model, using simulation only (no outcome data).
2. Choose $a \in (0, 1/2)$; we use $a = 1/4$ as a compromise. At $n = 16n_0$ it halves the radius ($16^{-1/4} = 1/2$), while $\sqrt{n}\theta_n$ still grows like $n^{1/4}$.
3. Report the implied radius and the implied worst-case local blind spot, of order θ_n (Theorems 3.6 and 3.7), at the actual sample size.

In our experiments $\theta_0 = 0.5$, $n_0 = 50$ and $a \in \{0.25, 0.45, 0.75\}$. These were fixed before any run and not changed afterwards.

4 Synthetic Experiments

4.1 Design

The kernel is Gaussian with fixed bandwidth $\sigma = 1$ ($\kappa = 1$). We use $\alpha = 0.05$, $B = 199$ permutations ($L = 200$), $B_f = 199$ fusion bootstrap draws, $B = 199$ partial-bootstrap draws, $Q_c = N(0, 1)$, and

sample sizes $n \in \{50, 100, 200, 400, 800\}$ with $m = n/2$ and $\ell_j = n$, so $\gamma = 1/3$ for one source. Normal laws are written $N(\text{mean}, \text{sd}^2)$.

Prespecified schedules. We use $\theta_0 = 0.5$ and $n_0 = 50$:

$$\theta_n^{\text{fix}} = 0.5, \qquad \theta_n^{(a)} = 0.5\,(n/50)^{-a}, \quad a \in \{0.25, 0.45, 0.75\},$$

so at $n = 800$ the radius is 0.25, 0.144 and 0.0625. Masses are $b_n^{(b)} = \frac{2}{3}(n/50)^{-b}$, $b \in \{0.25, 0.5\}$; $\frac{2}{3} = \ell/(m + \ell)$ is the full pooling weight at n_0. These constants and exponents were fixed before any simulation and were not changed afterwards. The exponent $a = 0.45$ lies inside the retention range $(0, 1/2)$ but close to its boundary; $a = 0.75$ lies outside it.

Fixed populations (B_n).

Setting	historical source(s)	treatment
NullIdent	$N(0, 1)$	$Q_t = Q_c$
NullShift	$N(0.3, 1)$ $(D = 0.131)$	$Q_t = Q_c$
NullFar	$N(1, 1)$ $(D = 0.421)$	$Q_t = Q_c$
PowIdent	$N(0, 1)$	$N(0.3, 1)$
BlindSpot	$N(0.8, 1)$ $(D = 0.342 < 0.5)$	$\frac{1}{3}N(0, 1) + \frac{2}{3}N(0.8, 1)$
Multi2Null	$N(0, 1)$ and $N(0.8, 1)$	$Q_t = Q_c$
Multi2Pow	$N(0, 1)$ and $N(0.8, 1)$	$N(0.3, 1)$

In BlindSpot the fixed radius eventually accepts the source, and the pooled control then equals Q_t (the population version of Theorem 3.7 with θ fixed). In the two-source settings the fusion level is $0.05/2$.

Sample-size-dependent populations (B_local). Let $\theta_n = \theta_n^{(0.25)}$.

Setting	historical source	treatment
BoundNull	$N(\mu_n, 1)$ with $D(Q_c, Q_{h,n}) = \theta_n$	$Q_t = Q_c$
LocBlind	$N(\mu_n, 1)$ with $D(Q_c, Q_{h,n}) = \theta_n/2$	$\frac{1}{3}Q_c + \frac{2}{3}Q_{h,n}$
LocPow	$N(0, 1)$	$N(2.5/\sqrt{n}, 1)$
LocAlign	$N(2/\sqrt{n}, 1)$	$N(2.5/\sqrt{n}, 1)$

LocBlind is the sequence of Theorem 3.7 with $c = 1/2$ for the $a = 0.25$ schedule. BoundNull places the source exactly on the shrinking boundary, where the selection never stabilises. LocPow and LocAlign are $\sqrt{n}$-local alternatives with an identical and a $\sqrt{n}$-shifted source, respectively.

Procedures. CUR; ALL; R-θ (reselected) and F-θ (frozen, partial permutation) for every schedule; PB-θ (frozen, original partial bootstrap) for θ^{fix} and $\theta^{(0.25)}$; PB-ALL (always merge, partial bootstrap; a direct check of the bootstrap at $Q_h = Q_c$); V-b for both masses; G (minP of CUR and R) for θ^{fix} and $\theta^{(0.25)}$. All procedures share data, permutations and bootstrap weights within a data set. The reselected tests use the label-invariant pooled-current bootstrap critical value of Section 3.1, and the frozen tests use the original bootstrap from the observed controls.

4.2 Monte Carlo protocol

There are 2000 data sets per null setting and 1000 per alternative, generated by NumPy's `SeedSequence` from `(20261018, setting, n, replicate)`. Confirmation runs with base seed 20261019 repeat both complete designs (1000/500 replicates for B_n and 2000/1000 for B_local). They were specified, as complete designs rather than selected cells, after the reproduction and Study A main results had been seen but before any Study B result had been seen. Wilson 95% intervals are shown as error bars, and in tables a bold null rate with ↑ marks an interval lying entirely above 0.05. The reproduction of the original Table 2 is shared with Study A.

5 Results

5.1 Reproduction of the original experiment

Table 2: Reproduction of Yang et al. (2026, Table 2, median-heuristic row), shared with Study A: merge rate and rejection rates, 1000 data sets per row ($m = 50$, $n = \ell = 100$, fixed $\theta = 0.4$). Bold with ↑: Wilson 95% interval above 0.05.

$\mu_c - \mu_t$	merge	Current only	PB (orig.)	PP frozen (orig.)	PP reselect	Always pool	minP guard
-0.4	0.772	0.490	0.402	0.539	0.459	0.457	0.482
+0.0	0.776	0.048	**0.078**↑	**0.067**↑	0.046	0.050	0.049
+0.4	0.789	0.523	0.731	0.653	0.598	0.664	0.568

Original Table 2 (median heuristic): merge 0.285/0.307/0.272; PB 0.577/0.064/0.572; PP 0.513/0.051/0.535.

Table 2 reproduces the qualitative behaviour of the original single-source procedure. Under the null the frozen partial bootstrap and frozen partial permutation reject at 0.078 and 0.067 (original: 0.064, 0.051); reselection gives 0.046. When the historical shift points in the direction of the effect ($\mu_c - \mu_t = -0.4$), the partial bootstrap is less powerful than current-only testing (0.402 vs. 0.490). This is the fixed-radius cancellation of Remark 3.19 that motivates this paper. Merge rates and power levels differ from the original. A check with the original fixed bandwidths (Table 6) brings the merge rates closer but not the power levels, and we could not identify the remaining difference. All our comparisons are therefore made within our own runs.

5.2 Type-I error with fixed populations

Table 3 and Fig. 1 show the following.

1. *Reselected radius tests, vanishing masses and the safeguard* stayed between 0.035 and 0.060 in all 20 null cells, as Proposition 3.1 guarantees. The only cells flagged are those at $n = 50$ in NullIdent. At $n = 50$ every schedule has the same radius, so these cells repeat one test on the same data, and current-only testing itself gave 0.059 there.
2. *Frozen selection with a shrinking radius is only pointwise valid.* With an identical source (NullIdent), F-$\theta^{(0.25)}$ starts at 0.076 and settles at 0.047–0.050 from $n = 200$ on, as Theorem 3.10 predicts. With a slightly shifted source (NullShift, $D = 0.131$) it stays at 0.064–0.074 up to $n = 800$. The source lies inside $\theta_n^{(0.25)}$ throughout the range (the radius only reaches 0.131 near $n \approx 10^4$), and its acceptance probability stays between 0.53 and 0.88, so the selection never stabilises. This is the non-uniformity anticipated in Section 3.5: the pointwise guarantee applies

Table 3: Null rejection rates (B_n, 2000 data sets per cell; $m = n/2$, $\ell_j = n$). Columns R are reselected, F frozen with partial permutation, PB frozen with the original partial bootstrap, PB all always merges with partial bootstrap, V vanishing mass, G minP safeguard. Bold with ↑: Wilson interval entirely above 0.05. At $n = 50$ all radius schedules coincide ($\theta = 0.5$).

Setting	n	CUR	R fix	R .25	R .45	R .75	F fix	F .25	PB fix	PB .25	PB all	V .25	V .50	G .25
NullIdent	50	0.059	**0.060**↑	**0.060**↑	**0.060**↑	**0.060**↑	**0.076**↑	**0.076**↑	**0.064**↑	**0.069**↑	0.034	0.048	0.048	0.057
	100	0.052	0.047	0.049	0.047	0.051	0.048	0.059	0.034	0.041	0.034	0.048	0.047	0.048
	200	0.040	0.048	0.048	0.042	0.040	0.048	0.049	0.028	0.031	0.029	0.048	0.047	0.043
	400	0.050	0.050	0.050	0.051	0.050	0.050	0.050	0.026	0.025	0.029	0.048	0.051	0.051
	800	0.052	0.047	0.047	0.051	0.052	0.047	0.047	0.024	0.024	0.025	0.048	0.044	0.050
NullShift	50	0.051	0.052	0.052	0.052	0.052	**0.071**↑	**0.071**↑	**0.085**↑	**0.089**↑	**0.064**↑	0.051	0.051	0.049
	100	0.052	0.047	0.055	0.057	0.053	**0.060**↑	**0.074**↑	**0.087**↑	**0.101**↑	**0.077**↑	0.049	0.049	0.050
	200	0.050	0.052	0.049	0.045	0.050	0.052	**0.071**↑	**0.096**↑	**0.110**↑	**0.092**↑	0.051	0.051	0.047
	400	0.051	0.050	0.051	0.051	0.051	0.050	**0.067**↑	**0.088**↑	**0.108**↑	**0.089**↑	0.049	0.050	0.044
	800	0.051	0.050	0.050	0.052	0.051	0.050	**0.064**↑	**0.086**↑	**0.096**↑	**0.081**↑	0.050	0.051	0.047
NullFar	50	0.046	0.045	0.045	0.045	0.045	0.046	0.046	0.048	0.048	**0.087**↑	0.046	0.046	0.046
	100	0.050	0.050	0.050	0.050	0.050	0.052	0.050	0.054	0.051	**0.075**↑	0.048	0.048	0.050
	200	0.058	0.057	0.058	0.058	0.058	**0.065**↑	0.058	**0.068**↑	0.058	**0.071**↑	0.057	0.057	0.058
	400	0.052	0.053	0.052	0.052	0.052	**0.073**↑	0.052	**0.078**↑	0.052	**0.063**↑	0.051	0.052	0.052
	800	0.049	0.042	0.049	0.049	0.049	**0.068**↑	0.049	**0.075**↑	0.049	0.051	0.044	0.044	0.049
Multi2Null	50	0.055	0.050	0.050	0.050	0.050	**0.073**↑	**0.073**↑	**0.081**↑	**0.077**↑	**0.107**↑	0.058	0.058	0.052
	100	0.035	0.047	0.045	0.035	0.036	**0.062**↑	0.057	**0.063**↑	0.041	**0.089**↑	0.044	0.042	0.037
	200	0.045	0.043	0.047	0.046	0.045	**0.064**↑	0.048	**0.109**↑	0.036	**0.098**↑	0.043	0.041	0.042
	400	0.056	0.047	0.057	0.054	0.056	**0.065**↑	0.057	**0.098**↑	0.032	**0.088**↑	0.047	0.046	0.053
	800	0.045	0.053	0.043	0.043	0.045	0.056	0.043	**0.082**↑	0.034	**0.079**↑	0.051	0.052	0.042

only once θ_n is well below every non-identical source distance. Before that, a frozen test can be miscalibrated for a long time.

3. *The original partial bootstrap.* With a fixed radius and a merged shifted source, PB-θ^{fix} rejected at 0.085–0.110 (NullShift, Multi2Null) and 0.068–0.078 (NullFar). With merged identical sources, the bootstrap is conservative: PB-ALL in NullIdent falls from 0.034 to 0.025 as n grows, and PB-$\theta^{(0.25)}$ in Multi2Null stays at 0.032–0.036 once only the identical source is merged. This supports Conjecture 3.13. The anti-conservative cases are nondegenerate ($Q_h \neq Q_c$), and there PB-ALL approaches 0.05 only slowly (NullFar: 0.087 → 0.051), which is consistent with the original asymptotic Theorem 3.5 but not with good finite-sample behaviour.

5.3 Power with fixed populations

The fixed-radius blind spot grows with n; shrinking radii remove it. In BlindSpot (Fig. 2; full table in Appendix B), the power of R-θ^{fix} falls from 0.546 ($n = 100$) to 0.000 ($n = 800$), because the source is accepted more and more often (0.17 → 1.00, Fig. 3). Current-only testing reaches 1.000. With $\theta^{(0.25)}$ the source is eventually rejected (0.002 acceptance at $n = 200$), and power follows the current-only curve with a lag (0.630, 0.821, 0.945, 0.999 against 0.738, 0.966, 1.000, 1.000). Faster schedules match current-only testing from $n = 200$. The frozen versions recover in the same way. Vanishing masses recover much more slowly ($b^{(0.25)}$: 0.647 at $n = 800$), as the prespecified mass shrinks only like $n^{-1/4}$.

Retaining identical sources pays off; shrinking too fast forfeits it. With an identical source (PowIdent), R-θ^{fix} and R-$\theta^{(0.25)}$ improve on current-only testing by 7–12 points for $100 \leq n \leq 400$ (e.g. 0.650 and 0.623 vs. 0.526 at $n = 200$), close to always pooling (0.650). With $a = 0.75$ the

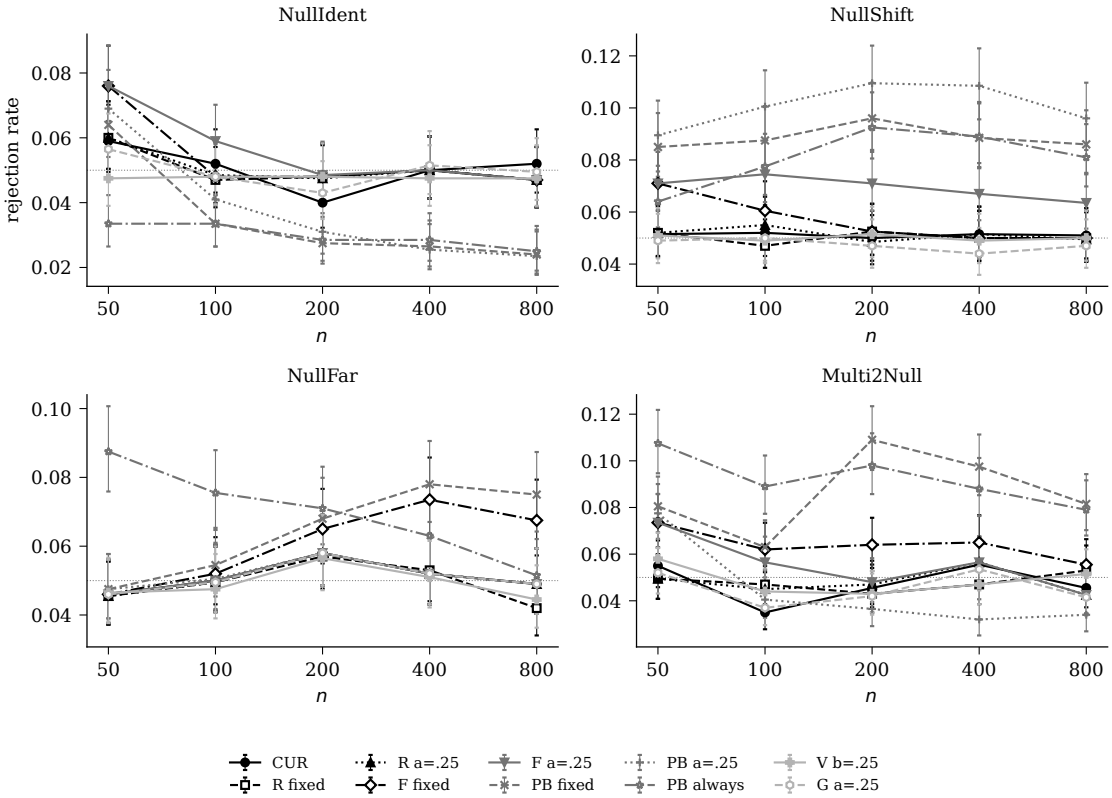


Figure 1: Null rejection rates against n (B_n), with Wilson 95% intervals and a dotted line at 0.05.

source is dropped from $n = 200$ on, as Theorem 3.3 predicts for $\sqrt{n}\theta_n \to 0$, and the test coincides with current-only testing. With $a = 0.45$, near the boundary $a = 1/2$, the source is accepted in 80–94% of data sets, yet reselection gains nothing (0.511 vs. 0.526 at $n = 200$). In this regime the selection in permuted rows is unstable, which adds noise to the reference distribution. At $n = 50$, all reselected tests lose to current-only testing (0.135 vs. 0.169); frozen tests appear better (0.233) but are not calibrated at that size.

Two sources (Multi2Pow). With one identical and one shifted source, the fixed radius eventually accepts both, and power collapses (0.014 at $n = 800$, against 0.989 for current-only testing). The treatment shift 0.3 points towards the shifted source, so pooling masks it. With $\theta^{(0.25)}$ only the identical source survives (Fig. 3), and power is above current-only testing for $n \geq 400$ (0.908 vs. 0.871; 0.996 vs. 0.989) and close to it below. This is the selection-stabilisation behaviour of Corollary 3.4.

5.4 Sample-size-dependent populations

The local blind spot (LocBlind). This is the sequence of Theorem 3.7 with $c = 1/2$. Current-only power rises from 0.255 to 0.779 between $n = 50$ and $n = 800$, since $\sqrt{n}\, D(Q_c, Q_{t,n})$ grows like $n^{1/4}$. R-$\theta^{(0.25)}$ follows it at first and then turns down: 0.311, 0.254, 0.176 and 0.071 at $n = 100, \ldots, 800$. Over the same range its acceptance rate in the observed row rises from 0.36 to 0.93. The fixed radius fails earlier (0.013 at $n = 800$), and so does ALL. The frozen F-$\theta^{(0.25)}$ and PB-$\theta^{(0.25)}$ behave like R-$\theta^{(0.25)}$ (0.085 and 0.072 at $n = 800$). The safeguard keeps most of the current-only power

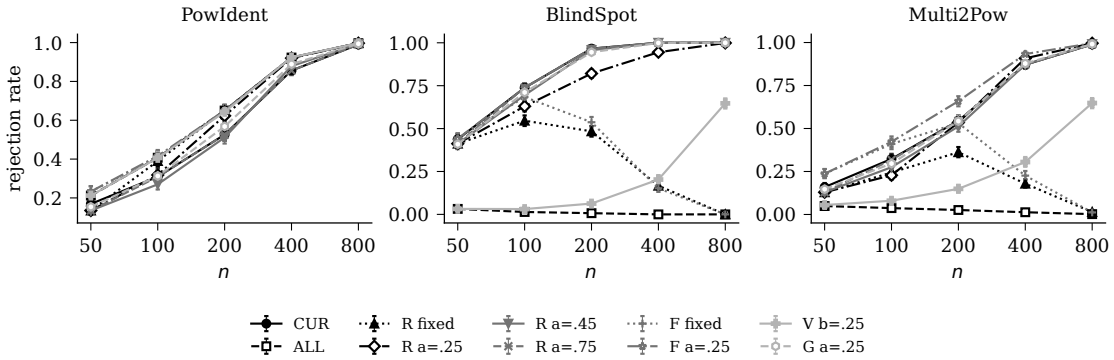


Figure 2: Power against n (B_n, 1000 data sets per point, Wilson 95% intervals). BlindSpot: $Q_h = N(0.8, 1)$, $Q_t = \frac{1}{3}N(0, 1) + \frac{2}{3}N(0.8, 1)$. Multi2Pow: sources $N(0, 1)$ and $N(0.8, 1)$, $Q_t = N(0.3, 1)$.

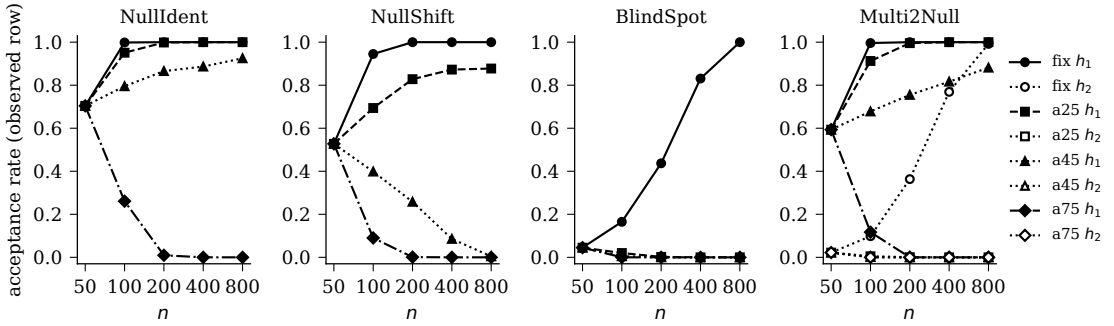


Figure 3: Acceptance rate of each source in the observed row, by schedule (B_n). In Multi2Null, $h_1 = N(0, 1)$ is identical and $h_2 = N(0.8, 1)$ is shifted.

(0.715). The faster schedules ($a = 0.45, 0.75$) escape this particular sequence only because their radius falls below $D(Q_c, Q_{h,n}) = \theta_n^{(0.25)}/2$, so they reject the source. A sequence built at their own radius would trap them in the same way. Vanishing masses recover slowly (0.234 and 0.580 at $n = 800$).

Identical source, $\sqrt{n}$-local alternative (LocPow). Retaining the identical source pays off here. For $n \geq 200$, R-θ^{fix} and R-$\theta^{(0.25)}$ reject at 0.29–0.31 against 0.22–0.23 for current-only testing, close to ALL (always pooling, which is optimal here). With $a = 0.75$ the source is discarded (acceptance 0.017 at $n = 200$), and the test coincides with current-only testing. The intermediate $a = 0.45$ retains the source in 81–92% of data sets but gains nothing; its rate (0.19–0.22) is even slightly below current-only testing. In reselected rows a source that is accepted only sometimes adds noise to the reference distribution. At $n = 50$, all reselected radius tests are below current-only testing (0.191 vs. 0.256), while frozen tests are above it (0.336); the frozen numbers are not comparable, because of size distortion at small n (Table 3). Vanishing masses (0.27–0.30) retain most of the gain over this range of n. This is consistent with Proposition 3.12, which is asymptotic: $b_n^{(0.25)}$ is still 0.33 at $n = 800$.

Shift aligned with the effect (LocAlign). When the historical source is shifted by $2/\sqrt{n}$ in the direction of a $2.5/\sqrt{n}$ effect, every procedure that keeps the source loses power (0.15–0.18

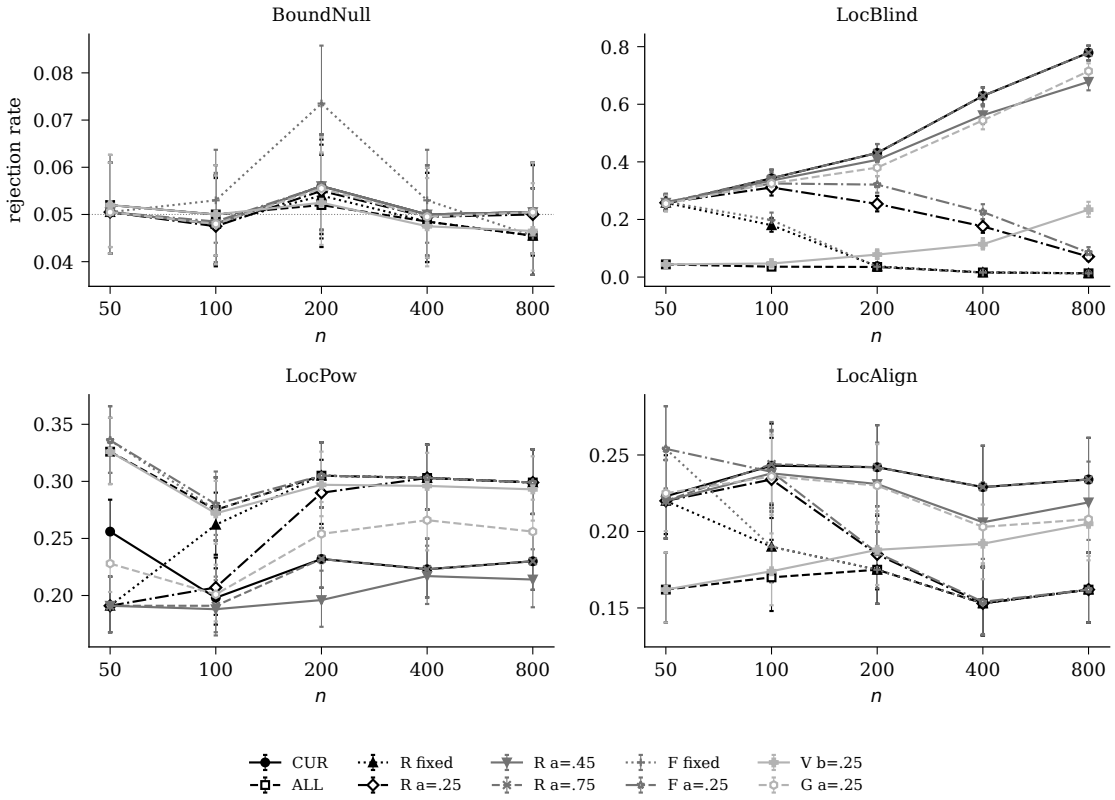


Figure 4: B_local: rejection rates against n when the populations change with n (Wilson 95% intervals; dotted line at 0.05 in BoundNull). The radius schedule used to construct BoundNull and LocBlind is $\theta_n^{(0.25)}$. Full table in Table 5.

against 0.23). This includes R-$\theta^{(0.25)}$ for $n \geq 400$, because a $\sqrt{n}$-local shift is always inside a radius $\gg n^{-1/2}$. The original partial bootstrap drops to 0.047–0.060, which is essentially no power. R-$\theta^{(0.45)}$ (0.21–0.23) and the safeguard (0.20–0.24) lose little. This is the $\sqrt{n}$-scale version of the trade-off in Corollary 3.8: a radius large enough to retain an identical source cannot distinguish it from a source shifted at the sampling scale.

Boundary null (BoundNull). With the source exactly at distance $\theta_n^{(0.25)}$, every reselected and fixed-rule test stays at 0.045–0.056. The $a = 0.25$ schedule almost never accepts the source (acceptance ≤ 0.002), since $\hat{D} + q$ exceeds θ_n whenever $\hat{D} \approx \theta_n$. The informative cell is the fixed radius. There the same source sits well inside $\theta = 0.5$ for large n, and its acceptance probability passes through intermediate values (0.36 at $n = 200$). In that cell the frozen test F-θ^{fix} rejects at 0.073 (interval above 0.05), while the reselected test gives 0.054. This is the predicted non-uniformity of frozen calibration: inflation appears exactly where the selection probability is neither near 0 nor near 1.

5.5 Independent-seed confirmation

Confirmation runs pending.

5.6 Computation

Per data set on one worker at $n = 50, 200, 800$: shared Gram matrix and permutation statistics 0.005, 0.026 and 0.29 s; label-invariant fusion bootstrap 0.008, 0.043 and 0.37 s; each additional radius schedule (reselected or frozen) under 1 ms, because all schedules reuse the same distances and critical values; partial bootstrap 0.003, 0.023 and 0.17 s. With a label-invariant critical value, reselection costs no more than freezing.

5.7 Discussion and conclusions

Answers to the questions posed.

- *When does power tend to one for every fixed alternative?* For the reselected test, whenever $\theta_n \to 0$ (Theorem 3.6). For the frozen test, under $\theta_n \to 0$ and $n\theta_n^2 \to \infty$, pointwise (Theorem 3.10). For vanishing mass, whenever $b_n \to 0$ (Proposition 3.11). The experiments confirm that the fixed-radius blind spot worsens with n (power 0.000 at $n = 800$) and that every shrinking schedule removes it.
- *How slowly must the radius shrink to keep an identical source?* $n\theta_n^2 \to \infty$ suffices, and $\sqrt{n}\theta_n \to 0$ loses it (Theorem 3.3). Empirically $a = 0.25$ kept the source (≥ 0.95 acceptance from $n = 100$), $a = 0.75$ lost it by $n = 200$, and $a = 0.45$ kept it in 80–94% of data sets without any power benefit.
- *Too fast or too slow.* Too fast forfeits borrowing (PowIdent, $a = 0.75$). Too slow prolongs the blind spot (BlindSpot, $a = 0.25$ lags current-only testing up to $n = 400$). It also prolongs the miscalibration of frozen tests (NullShift) and leaves a local blind spot of order θ_n (LocBlind, power 0.071 at $n = 800$ against 0.779). No schedule avoids all three (Corollary 3.8).
- *Local and sample-size-dependent discrepancies.* A source at $\sqrt{n}$-local distance is indistinguishable from an identical one for every schedule that retains identical sources, so its effect on local power is inherited unchanged (LocAlign). Near the radius, the selection stays random and frozen tests are miscalibrated (BoundNull with fixed θ).

Recommendations. If a radius schedule is used, combine it with full reselection and a label-invariant fusion critical value. Calibration is then exact and costs nothing extra. Choose a well inside $(0, 1/2)$ and report the implied worst-case local blind spot. When uniform protection at the $\sqrt{n}$ scale matters, add the minP safeguard. In our experiments G-$\theta^{(0.25)}$ was never more than 0.07 below current-only testing in any setting (the worst case was LocBlind at $n = 800$, 0.715 against 0.779). It kept part of the borrowing gain with an identical source (0.568 vs. 0.526 at $n = 200$). Do not rely on the frozen partial bootstrap: it was anti-conservative with shifted merged sources and conservative with identical ones.

What remains open. We did not prove that any schedule is optimal, and we expect none to be: the trade-off of Corollary 3.8 suggests that the right object is a rule that is soft at the $n^{-1/2}$ scale (as in Study A), not a better exponent. Further open questions are the limit of the acceptance probability at $a = 1/2$; a proof or disproof of Conjecture 3.13 for general kernels; local power expansions for reselected tests; and data-adaptive radii with a valid selection correction. Our simulations use one-dimensional Gaussian families and one kernel bandwidth. The sizes of the effects we report, though not their direction, should be expected to change with the kernel and the dimension.

A Proofs

Proofs of A-Theorems 3.2 and 3.13 and A-Proposition 3.16 are in Study A, Appendix A. We repeat the two facts that are used repeatedly. (F1) For an i.i.d. sample of size M from R, $\mathbb{E}\|\hat{\xi} - \xi_R\|^2 = v_R/M \le \kappa^2/M$. (F2) For a uniform m-subset A of fixed pooled values, $\mathbb{E}_A\|\hat{\xi}_{c,A} - \hat{\xi}_{t,A}\|^2 \le M_{m,n}$, and since $\hat{\xi}_{c,A} - \hat{\xi}_{\text{pool}} = \eta(\hat{\xi}_{c,A} - \hat{\xi}_{t,A})$, also $\mathbb{E}_A\|\hat{\xi}_{c,A} - \hat{\xi}_{\text{pool}}\|^2 \le \eta^2 M_{m,n}$.

Proof of Lemma 3.2. As in A-Lemma 3.10. For the pooled variant, $\mathbb{E}^*\|\hat{\xi}^*_{\text{pool}} - \hat{\xi}_{\text{pool}}\|^2 \le \kappa^2/m$ by (F1) applied to the bootstrap distribution with $M = m$ draws. □

Proof of Theorem 3.3. Write $\epsilon_j = \|(\hat{\xi}_j - \hat{\xi}_c) - (\xi_j - \xi_c)\|$, so $\mathbb{E}\epsilon_j^2 = v_c/m + v_j/\ell_j \le s_j$ by (F1) and independence, and $|\hat{D}_j - d_j| \le \epsilon_j$. (a) Exclusion requires $\hat{D}_j + q_j \ge \theta_n$, hence $\epsilon_j + q_j \ge \theta_n - d_j$, hence $\epsilon_j \ge (\theta_n - d_j)/2$ or $q_j \ge (\theta_n - d_j)/2$. Apply Markov's inequality to ϵ_j^2 and q_j^2 (Lemma 3.2). (b) Inclusion requires $\hat{D}_j < \theta_n$, hence $\epsilon_j > d_j - \theta_n$, and Markov's inequality gives the bound. In a permutation row the control sample is a uniform subset of the pool. Replace ϵ_j by $\|(\hat{\xi}_j - \hat{\xi}_{c,A}) - (\xi_j - \xi_d)\|$ and d_j by $D(Q_d, Q_{h_j})$ with $\xi_d = \lambda\xi_c + (1-\lambda)\xi_t$; (F1), (F2) and the triangle inequality give the same bounds with s_j replaced by $3(s_j + \eta^2 M_{m,n})$. For the consequences: if $d_j > 0$ is fixed and $\theta_n \to 0$, (b) gives $O(n^{-1})$. If $d_j = 0$, (a) gives $O(1/(n\theta_n^2))$, using $\bar{q}_{n,j}^2 = O(n^{-1})$. If $\sqrt{n}\theta_n \to 0$ and $d_j = 0$, inclusion requires $\sqrt{n}\hat{D}_j < \sqrt{n}\theta_n \to 0$. But $\sqrt{n}(\hat{\xi}_j - \hat{\xi}_c)$ converges in distribution to a centred Gaussian element G in $\mathcal{H}_k$ with covariance $(c_1^{-1} + c_{2j}^{-1})\Sigma$ (Hilbert-space CLT; see Yang et al., 2026), and $\Sigma \ne 0$ when $v_{Q_c} > 0$. A nondegenerate centred Gaussian measure on a separable Hilbert space gives mass zero to $\{0\}$, so by the portmanteau theorem $\limsup \mathbb{P}(\sqrt{n}\hat{D}_j < \varepsilon) \le \mathbb{P}(\|G\| \le \varepsilon) \to 0$ as $\varepsilon \downarrow 0$. The triangular-array statements follow from (a) and (b), whose right-hand sides depend on the distributions only through $d_{j,n}$ and κ. □

Proof of Corollary 3.4. A union bound over j of the two cases of Theorem 3.3. □

Proof of Theorem 3.6. In every row each accepted source satisfies $D(Q^m_{c,A_b}, Q^{\ell_j}_{h_j}) < \theta_n - q_j \le \theta_n$, so with sample-size pooling $D(\hat{Q}_{f,b}, Q^m_{c,A_b}) \le \rho\theta_n$ (A-Lemma 3.9). Apply A-Theorem 3.12 with $\delta = \rho\theta_n$. Its bound depends on the distributions only through d and κ, which gives the uniform statement. □

Proof of Theorem 3.7. Write $\theta = \theta_n$, $\xi_h = \xi_{Q_{h,n}}$, $\xi_t = \xi_{Q_{t,n}} = \gamma\xi_c + (1-\gamma)\xi_h$, so $\|\xi_t - \xi_c\| = (1-\gamma)c\theta$ and $\sqrt{n}\|\xi_t - \xi_c\| \to \infty$. The current-only test and G are consistent along this sequence by A-Theorem 3.12 with $\delta = 0$ (and A-Proposition 3.16 for G), because the bound depends only on d_n and $\sqrt{n}d_n \to \infty$.

Observed row. $\hat{D} \le c\theta + \epsilon$ with $\mathbb{E}\epsilon^2 \le s = O(n^{-1})$, and $q = O_P(n^{-1/2})$. Since $\sqrt{n}\theta \to \infty$, $\mathbb{P}(\hat{D} + q < \theta) \ge 1 - \mathbb{P}(\epsilon + q \ge (1-c)\theta) \to 1$: the source is accepted. Then $\hat{\xi}_f - \hat{\xi}_t = \gamma e_c + (1-\gamma)e_h - e_t$, where $e_\cdot$ are the centred empirical errors, because $\gamma\xi_c + (1-\gamma)\xi_h - \xi_t = 0$. So $\mathbb{E}S_0 \le \kappa^2(\gamma^2/m + (1-\gamma)^2/\ell + 1/n) = O(n^{-1})$ and $nS_0 = O_P(1)$.

Rows $b \ge 1$. Let $\xi_d = \lambda\xi_c + (1-\lambda)\xi_t$. By (F1) and (F2), $\hat{\xi}_{c,A_b} - \xi_d = O_P(n^{-1/2})$ and $\hat{\xi}_{t,A_b} - \xi_d = O_P(n^{-1/2})$. Now $\xi_d - \xi_h = \{\lambda + (1-\lambda)\gamma\}(\xi_c - \xi_h)$, so $D(Q^m_{c,A_b}, Q^\ell_h) \le \kappa_0 c\theta + O_P(n^{-1/2})$ with $\kappa_0 = \lambda + (1-\lambda)\gamma < 1$, and the source is accepted in every row with probability tending to one. Then

$$\hat{\xi}_{f,b} - \hat{\xi}_{t,A_b} = (1-\gamma)(\xi_h - \xi_d) + O_P(n^{-1/2}), \qquad \|(1-\gamma)(\xi_h - \xi_d)\| = (1-\gamma)\kappa_0 c\theta,$$

so $\sqrt{nS_b} \geq \sqrt{n}(1-\gamma)\kappa_0 c\theta - O_P(1) \to \infty$. With B fixed, $\mathbb{P}(S_b > S_0 \; \forall b \geq 1) \to 1$, on which event $p = 1$. The frozen test uses the observed (accepted) selection in every row, and the same argument applies. □

Remark A.1 (Partial bootstrap along the same sequence). For PB-θ_n, $\|\hat{\xi}_f - \hat{\xi}_c\|^2 \approx (1-\gamma)^2 c^2 \theta^2$ and $S_0 = O_P(n^{-1})$, so $\Delta \approx -\sqrt{n}(1-\gamma)^2 c^2 \theta^2$, whereas the bootstrap statistic is $O_P(\theta) + O_P(n^{-1/2})$. Since $\sqrt{n}\theta^2/\theta = \sqrt{n}\theta \to \infty$, Δ lies below the bootstrap quantile with probability tending to one. We have not written out the conditional-law details.

Proof of Theorem 3.10. Use the same permutations for both tests. On $\{\hat{S} = S_0\}$ the frozen test and the oracle coincide, so $|\mathbb{P}(\text{F rejects}) - \mathbb{P}(\phi^{\text{or}} = 1)| \leq \mathbb{P}(\hat{S} \neq S_0) \leq e_n$ (Corollary 3.4). The oracle is a fixed-rule partial-permutation test and has level $\lfloor \alpha L \rfloor / L$. Under a fixed alternative, the oracle pools only sources distributed as Q_c, so its observed fused embedding converges to ξ_c and $S_0 \to d^2 = D^2(Q_c, Q_t)$. In row $b \geq 1$ the fused embedding converges to $\gamma_0 \xi_d + (1-\gamma_0)\xi_c$ with $\gamma_0 = m/(m + \ell_{S_0})$, and $S_b \to \|(1-\gamma_0)(\xi_c - \xi_d)\|^2 = (1-\gamma_0)^2(1-\lambda)^2 d^2 < d^2$ (with $\gamma_0 = 1$ if $S_0 = \emptyset$). With B fixed, $p \to 1/L \leq \alpha$ in probability, so the oracle is consistent, and so is the frozen test. □

Proof of Proposition 3.11. The weights are fixed, so A-Theorem 3.2 applies with a constant rule. In every row $\hat{\xi}_{f,b} - \hat{\xi}_{c,A_b} = \sum_j w_{n,j}(\hat{\xi}_j - \hat{\xi}_{c,A_b})$, whose norm is at most $b_n \max_j \|\hat{\xi}_j - \hat{\xi}_{c,A_b}\| \leq 2\kappa b_n$. Then apply A-Theorem 3.12 with $\delta = 2\kappa b_n$. □

Proof of Proposition 3.12. With $\Delta_b = \hat{\xi}_{c,A_b} - \hat{\xi}_{t,A_b}$ we have $S_b^{\text{cur}} = \|\Delta_b\|^2$ and $S_b^{\text{bor}} = \|\Delta_b + u_b\|^2$, so $|S_b^{\text{bor}} - S_b^{\text{cur}}| \leq 2\|\Delta_b\|\|u_b\| + \|u_b\|^2$. Under local alternatives $nS_b^{\text{cur}} = O_P(1)$ in every row, by (F1) for $b = 0$ and (F2) for $b \geq 1$. If all sources equal Q_c, then $\hat{\xi}_j - \hat{\xi}_{c,A_b} = O_P(n^{-1/2})$, because $\xi_j = \xi_c$ and $\xi_d - \xi_c = (1-\lambda)(\xi_{t,n} - \xi_c) = O(n^{-1/2})$; hence $\sqrt{n}\|u_b\| = O_P(b_n)$. If $d_j > 0$ is fixed, $\|u_b\| = O_P(b_n)$, and $\sqrt{n}b_n \to 0$ makes both terms $o_P(n^{-1})$. □

B Additional results

Table 4: Power with fixed populations (B_n, 1000 data sets per cell). G fix and G .25 are minP combinations of current-only testing with R-θ^{fix} and R-$\theta^{(0.25)}$.

Setting	n	CUR	ALL	R fix	R .25	R .45	R .75	F .25	PB .25	V .25	V .50	G .25	G fix
PowIdent	50	0.169	0.212	0.135	0.135	0.135	0.135	0.233	0.245	0.212	0.212	0.154	0.154
	100	0.309	0.407	0.390	0.318	0.270	0.309	0.416	0.371	0.406	0.404	0.313	0.347
	200	0.526	0.650	0.650	0.623	0.511	0.525	0.650	0.639	0.643	0.638	0.568	0.583
	400	0.856	0.923	0.923	0.920	0.879	0.856	0.923	0.921	0.924	0.920	0.889	0.891
	800	0.990	0.997	0.997	0.997	0.993	0.990	0.997	0.996	0.997	0.996	0.994	0.994
BlindSpot	50	0.438	0.032	0.412	0.412	0.412	0.412	0.441	0.437	0.032	0.032	0.409	0.409
	100	0.738	0.014	0.546	0.630	0.695	0.739	0.737	0.736	0.031	0.066	0.711	0.699
	200	0.966	0.007	0.484	0.821	0.957	0.966	0.965	0.965	0.063	0.311	0.945	0.941
	400	1.000	0.000	0.167	0.945	1.000	1.000	1.000	1.000	0.203	0.867	0.999	0.999
	800	1.000	0.000	0.000	0.999	1.000	1.000	1.000	1.000	0.647	1.000	1.000	1.000
Multi2Pow	50	0.159	0.049	0.129	0.129	0.129	0.129	0.237	0.244	0.055	0.055	0.147	0.147
	100	0.323	0.037	0.244	0.228	0.275	0.318	0.423	0.386	0.080	0.111	0.296	0.296
	200	0.545	0.026	0.362	0.535	0.512	0.545	0.659	0.659	0.149	0.267	0.543	0.503
	400	0.871	0.013	0.179	0.908	0.872	0.871	0.933	0.939	0.305	0.632	0.878	0.813
	800	0.989	0.002	0.014	0.996	0.991	0.989	0.996	0.997	0.648	0.944	0.991	0.984

Table 5: Sample-size-dependent populations (B_local; BoundNull 2000 data sets per cell, others 1000). Bold with ↑: Wilson interval entirely above 0.05 (BoundNull only).

Setting	n	CUR	ALL	R fix	R .25	R .45	R .75	F fix	F .25	PB .25	V .25	V .50	G .25
BoundNull	50	0.051	0.052	0.051	0.051	0.051	0.051	0.051	0.051	0.051	0.052	0.052	0.051
	100	0.048	0.050	0.048	0.048	0.048	0.048	0.053	0.049	0.049	0.050	0.050	0.048
	200	0.056	0.052	0.054	0.055	0.056	0.056	**0.073**$^{\uparrow}$	0.056	0.056	0.052	0.056	0.056
	400	0.050	0.049	0.049	0.050	0.050	0.050	0.053	0.050	0.051	0.048	0.048	0.050
	800	0.051	0.045	0.045	0.050	0.051	0.051	0.045	0.051	0.051	0.046	0.046	0.051
LocBlind	50	0.255	0.044	0.258	0.258	0.258	0.258	0.261	0.261	0.255	0.044	0.044	0.255
	100	0.343	0.036	0.180	0.311	0.335	0.343	0.198	0.325	0.312	0.047	0.063	0.325
	200	0.431	0.035	0.037	0.254	0.407	0.431	0.036	0.321	0.294	0.078	0.151	0.380
	400	0.629	0.016	0.016	0.176	0.562	0.629	0.016	0.226	0.211	0.114	0.329	0.544
	800	0.779	0.013	0.013	0.071	0.678	0.779	0.013	0.085	0.072	0.234	0.580	0.715
LocPow	50	0.256	0.326	0.191	0.191	0.191	0.191	0.336	0.336	0.324	0.326	0.326	0.228
	100	0.198	0.275	0.262	0.207	0.188	0.191	0.275	0.280	0.264	0.272	0.279	0.201
	200	0.232	0.305	0.305	0.290	0.196	0.232	0.305	0.305	0.251	0.297	0.289	0.254
	400	0.223	0.303	0.303	0.303	0.217	0.223	0.303	0.303	0.256	0.296	0.286	0.266
	800	0.230	0.299	0.299	0.299	0.214	0.230	0.299	0.299	0.271	0.293	0.271	0.256
LocAlign	50	0.223	0.162	0.220	0.220	0.220	0.220	0.254	0.254	0.221	0.162	0.162	0.225
	100	0.243	0.170	0.190	0.234	0.238	0.244	0.190	0.239	0.146	0.174	0.187	0.236
	200	0.242	0.175	0.175	0.185	0.231	0.242	0.175	0.186	0.060	0.188	0.205	0.230
	400	0.229	0.153	0.153	0.153	0.206	0.229	0.153	0.154	0.047	0.192	0.204	0.203
	800	0.234	0.162	0.162	0.162	0.219	0.234	0.162	0.162	0.058	0.205	0.219	0.208

Table 6: Original Table 2 rows with fixed bandwidth s, read as $k(x, y) = e^{-(x-y)^2/s^2}$; 1000 data sets per cell; frozen selection as in the original. "original" rows copy Yang et al. (2026, Table 2).

s	merge	$\mu_c - \mu_t = -0.4$		$\mu_c = \mu_t$		$\mu_c - \mu_t = +0.4$	
		PB	PP	PB	PP	PB	PP
0.5	0.235	0.338	0.358	0.055	0.059	0.398	0.386
original	0.34	0.528	0.511	0.070	0.063	0.536	0.566
1.0	0.598	0.407	0.504	0.062	0.058	0.626	0.569
original	0.67	0.572	0.559	0.074	0.053	0.711	0.629
1.5	0.806	0.398	0.584	**0.088**$^{\uparrow}$	**0.064**$^{\uparrow}$	0.694	0.625
original	0.82	0.581	0.566	0.074	0.065	0.739	0.677

C Implementation and reproducibility

Code. All code is in the repository `https://github.com/Yu-Jusang/ttp-extensions` (directory `code/`) and in the accompanying archive.

- `ttp.py`: kernel, Gram-matrix bookkeeping, allocations, row statistics, p-values (ties counted, observed row included), minP calibration, risk-weight selection, equivalence-fusion bootstrap (row-wise with value-sorted common weights, or label-invariant pooled), partial bootstrap.
- `methods.py`: the procedures of Study A (`study_A`) and Study B (`study_B`), sharing data, permutations and bootstrap weights within a data set.
- `run_sims.py`: suites `repro`, `repro_bw`, `A_main`, `A_n`, `B_n`, `B_local`; `run_all.sh` (exploratory, base seed 20261018) and `run_conf.sh` (confirmation, base seed 20261019).
- `analyze.py`: all tables (LaTeX fragments in `paper_*/gen/`) and figures from the raw arrays.
- `pb_scalar.py`: the Gaussian-limit calculation for the partial bootstrap at degeneracy (Study B).

How to run. From the repository root:

```
cd code
bash run_all.sh
bash run_conf.sh
python analyze.py
```

Each suite writes `results/<suite>.npz`, which holds one array of p-values (or 0/1 decisions for partial-bootstrap procedures), timings and diagnostics per setting and procedure. It also writes a metadata file `results/<suite>_meta.json` recording the seed, replicate counts, elapsed time and software versions. Every data set is regenerated by NumPy's `SeedSequence` from `(seed, setting, level, replicate)`.

Environment. Windows 11, Intel Core i7-1360P (12 cores, 16 threads; 15 worker processes, one BLAS thread each), Python 3.12.2, NumPy 2.5.4, SciPy 1.18.1, Matplotlib 3.11.2, TeX Live 2024. Reported times are per data set on one worker and include only the step named.

Implementation details. (i) All statistics are quadratic forms in one Gram matrix K of the $N+\sum_j \ell_j$ outcomes. A permutation row is a pair of weight vectors $(\mathbf{1}_A/m, \mathbf{1}_{A^c}/n)$, and all L rows are processed by one matrix product. (ii) Rows $b \geq 1$ are uniform m-subsets drawn with replacement across rows; the observed allocation is row 0. (iii) Ties: $p = L^{-1}\#\{b : S_b \geq S_0 - 10^{-12}\max(1, |S_0|)\}$. (iv) The fusion quantile is the $\lceil(1-\alpha_f)B_f\rceil$-th order statistic. (v) The multinomial bootstrap weights of the current control are attached to the controls sorted by value (the symmetry assumption, Assumption 3.1 of Study A). (vi) The risk-weight grid contains 0 and is sorted by total weight, so ties are broken towards less borrowing.

What was not done. We did not analyse real data. The Prospera application of the original paper was not repeated because its data are not distributed with the paper, and the original code was not available to us. Differences from the original Table 2 are discussed in the reproduction section.

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